

Vedant Desai

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EDUCATION

University of Michigan

B.S. in Computer Science and Economics

Expected May 2028

Ann Arbor, MI

- **GPA:** 3.7 / 4.0

- **Coursework:** Data Structures & Algorithms, Linear Algebra for Machine Learning, Microeconomics, Macroeconomics, Discrete Mathematics, Calculus III, Physics (Mechanics)

EXPERIENCE

Vylet

May 2026 – Present | vyletdata.com

Founder — Automated lead-sourcing platform for PE/search-fund; live product generating \$1,500 MRR across three clients

- Launched Vylet, automating a ~30-minute manual process per business into a Dockerized LangGraph pipeline generating 30 scored leads in 30 minutes — a 30x speedup — with Redis/Celery workers on a recurring cycle.
- Engineered a LangSmith eval pipeline spanning 20 adversarial business test cases across 13 archetype labels — manufacturers, SaaS tools, PE holding companies, geographic mismatches — then layered deterministic Pydantic consensus gates to lift extraction faithfulness from 50% to 90%.

Lyndbrook Capital

Feb 2026 – Apr 2026

Data Engineering Consultant

Remote

- Contracted pre-LOI to build acquisition intelligence for a boutique search fund targeting water utility operators; aggregated EPA ECHO and MassGIS regulatory data into a unified PWSID entity database and delivered 800+ validated Day-1 acquisition targets within the engagement window.
- Built a Review Velocity scoring algorithm that proxied fleet expansion and operational scale from public compliance data, filtering 800 acquisition targets down to a 280-lead qualified shortlist — a 35% precision rate against the fund's revenue criteria.

CaseStudyPrep.AI

Dec 2025 – May 2026

Software Engineer Co-op (Voice AI)

Remote

- Eliminated the silent-audio problem in a voice-AI product — most frames sent to Whisper were dead air — by running Silero VAD client-side via ONNX Runtime, filtering silence before upload and cutting cloud inference costs by 40%.
- Eliminated a 27% audio upload failure rate with fault-tolerant RxJS logic that detects expired S3 presigned URLs, regenerates them mid-flight, and negotiates MIME types for WAV files Angular silently rejected.

PROJECTS

SignalWeaver | *Python, LoRA, FastAPI*

github.com/Verdent06/SignalWeaver

- Lifted financial-sentiment classification accuracy from 81% to 96% by LoRA fine-tuning a quantized meta-llama/Meta-Llama-3.1-8B-Instruct model on 3,454 Financial PhraseBank entries, evaluated on a held-out test set.
- Served composite financial-research scores through async REST endpoints (FastAPI) wrapping fundamentals, MPNet sentiment, and regression logic, instrumented end-to-end at 9.1s p50 / 15.2s p99 across 90 successful runs on 90 tickers.

Granular Synthesizer Plugin | *C++, JUCE, VST3, AU*

github.com/Verdent06/granular-synth

- Enforced the audio thread's zero-allocation constraint by pre-allocating a `MemoryPool<Grain, 64>` slab per voice at plugin startup — a fixed free-list that acquires and releases grain slots without touching the heap — so `processBlock()` never calls `new` or `delete` after `prepareToPlay()` completes.
- Configured CMake to compile VST3 and AU bundles from a single JUCE codebase — targeting a macOS universal binary (arm64 + x86_64) for native Apple Silicon and Intel — and audited every `processBlock()` path against a real-time safety checklist confirming zero heap allocations and zero lock acquisitions before release builds.

TECHNICAL SKILLS

Languages Python, TypeScript, C++, SQL

AI / ML LangGraph, LangSmith, ONNX Runtime, Silero VAD

Frameworks FastAPI, Angular, RxJS, JUCE, Pydantic

Databases Redis

Cloud & Infrastructure Docker, Celery, Git